UNIT - II

PRODUCTION AND COST ANALYSIS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Production: nature, meaning, significance, functions, advantages
- Production function, least-cost combination, short run and long run
- Isoquants and isocosts
- Cost concepts and cost behaviour
- Break-Even Analysis (BEA) and break-even point (simple problems)

1. Introduction to Production

1.1 Meaning

- **Production** = creation of utility; transforming inputs (land, labour, capital, organisation) into outputs (goods and services).
- **Production analysis** studies the relation between inputs and output to find the most efficient way of producing.

Production is any activity that creates or adds value (utility): form utility, place utility, time utility and possession utility.

1.2 Nature

- Production is a **process** of converting inputs into output.
- It is **technical** (depends on technology) and **economic** (aims at minimum cost).
- It involves **factors of production**: Land (rent), Labour (wages), Capital (interest), Organisation / Entrepreneur (profit).
- Output can be **tangible** (car) or **intangible** (service).

1.3 Significance and functions

- Helps determine **what, how and for whom** to produce.
- Helps in **optimum utilisation of resources** and cost reduction.

- Basis for **planning** capacity, labour, raw material and inventory.
- Helps in **pricing** and **profit** decisions and in choosing technology.

1.4 Advantages of production analysis

- Finds the **least-cost** input combination, so efficiency improves.
- Helps the manager decide whether to expand output in the short or long run.
- Identifies **returns to scale** and economies of scale.
- Supports **capital budgeting** and make-or-buy decisions.

2. Production Function

2.1 Meaning

The production function shows the **technical relationship between physical inputs and the maximum output** obtainable, given the state of technology.

$$Q = f(L, K, N, T)$$

- Q = output; L = labour; K = capital; N = land / raw material; T = technology.
- Cobb-Douglas production function: **Q = A L^a K^b** . If a + b = 1, constant returns to scale.

2.2 Short run vs Long run

Basis	Short run	Long run
Inputs	At least one factor fixed (capital)	All factors variable
Law	Law of variable proportions	Returns to scale
Time	Not enough to change plant size	Enough to change plant, enter / exit
Costs	Fixed and variable costs	All costs variable

2.3 Short-run production: Law of Variable Proportions

Statement: When more and more units of a variable factor (labour) are combined with fixed factors, the **total output initially increases at an increasing rate, then at a decreasing rate, and finally falls** . (Law of diminishing returns)

- **TP** = total product; **AP** = TP / L; **MP** = change in TP / change in L.

Illustration (land fixed = 1 acre)

Labour	TP	AP	MP	Stage
1	10	10	10	Stage I: increasing returns

2	24	12	14	Stage I
3	36	12	12	Stage II begins (MP falls)
4	44	11	8	Stage II: diminishing returns
5	48	9.6	4	Stage II
6	48	8	0	End of Stage II (TP max)
7	42	6	-6	Stage III: negative returns

- **Stage I** : MP rises, AP rises; **Stage II** : MP falls but positive; the rational firm operates here; **Stage III** : MP negative, TP falls.
- Relationship: MP cuts AP at the maximum point of AP; when MP = 0, TP is maximum.

Assumptions

- Technology is constant; one factor variable, others fixed; units of the variable factor are homogeneous.

2.4 Long run: Returns to Scale

- Increase **all** inputs in the same proportion; see the effect on output.

Type	Meaning	Example
Increasing returns	Output rises more than proportionately	Inputs x2, output x3
Constant returns	Output rises in the same proportion	Inputs x2, output x2
Decreasing returns	Output rises less than proportionately	Inputs x2, output x1.5

- Causes: **economies of scale** (specialisation, bulk buying) give increasing returns; **diseconomies** (management problems) give decreasing returns.

3. Isoquants and Isocosts

3.1 Isoquant (equal product curve)

An isoquant is a curve showing **all combinations of two inputs (labour and capital) that yield the same level of output**.

Properties of isoquants

- Slopes **downward** from left to right (if L rises, K must fall).
- **Convex to the origin** because the MRTS diminishes.
- Two isoquants **never intersect**.
- Higher isoquant = higher level of output.

- They do not touch either axis (both inputs are needed).

MRTS (L for K) = - dK / dL = MPL / MPK

- **MRTS** (Marginal Rate of Technical Substitution) = the amount of capital given up to get one more unit of labour, keeping output constant.

Types of isoquant shapes

Shape	Meaning
Convex (usual)	Inputs imperfect substitutes
Straight line	Inputs are perfect substitutes
L-shaped	Fixed proportion inputs (perfect complements)

3.2 Isocost line

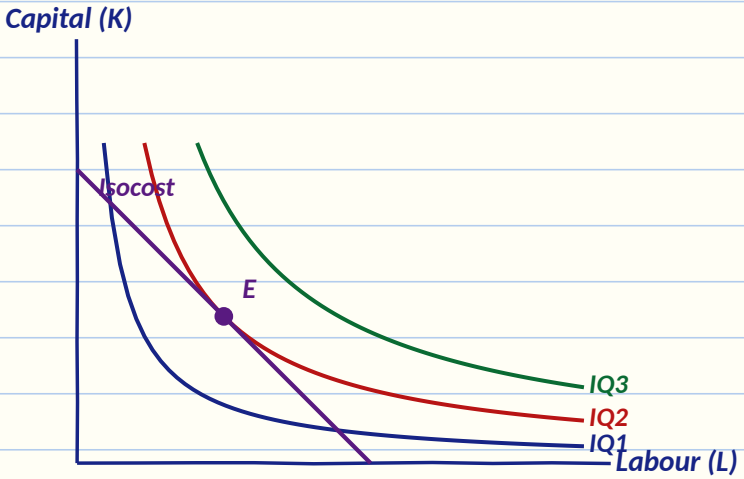
The isocost line shows **all combinations of two inputs that can be purchased for a given total outlay (cost)** at given input prices.

C = wL + rK

- w = wage rate, r = price of capital; slope of isocost = w / r; intercepts: C/w on L-axis and C/r on K-axis.

Example

Total budget = Rs. 1000; w = Rs. 20; r = Rs. 50. Maximum labour = 1000/20 = 50 units; maximum capital = 1000/50 = 20 units.



Isoquant: same output, different input mixes.

Isocost: same total cost.

Least-cost point E = tangency (MRTS = w/r)

3.3 Least-cost combination (producer's equilibrium)

- The firm chooses the input mix that gives a given output at the **minimum cost** (or maximum output for a given cost).

Condition: MRTS = MPL / MPK = w / r

- Graphically it is the **tangency point** of the isoquant and the isocost line; the isoquant must be convex at that point.
- Equi-marginal form: **MPL / w = MPK / r** (the last rupee spent on each factor gives the same extra output).

Expansion path

- Locus of the tangency points when the budget rises with constant prices. It shows how the firm expands output along least-cost combinations.

4. Cost Concepts

4.1 Meaning

- Cost = the expenditure incurred in producing a good (sacrifice made in terms of money).

4.2 Important cost concepts

Cost	Meaning
Explicit (actual) cost	Money paid to outsiders: wages, rent, raw material
Implicit cost	Imputed cost of owner's own resources (own capital, own time)
Opportunity cost	Value of the best alternative forgone
Fixed cost (TFC)	Does not change with output: rent, insurance, salaries
Variable cost (TVC)	Changes with output: raw material, piece wages
Total cost	TC = TFC + TVC
Sunk cost	Already spent; cannot be recovered; ignore in decisions
Out-of-pocket cost	Current cash payments (opposite: book cost like depreciation)
Incremental cost	Change in total cost due to a decision (e.g. new product)
Marginal cost	Extra cost of producing one more unit
Private vs Social cost	Cost to the firm vs cost to society (pollution)

4.3 Cost formulas

AFC = TFC / Q ; AVC = TVC / Q ; AC = TC / Q

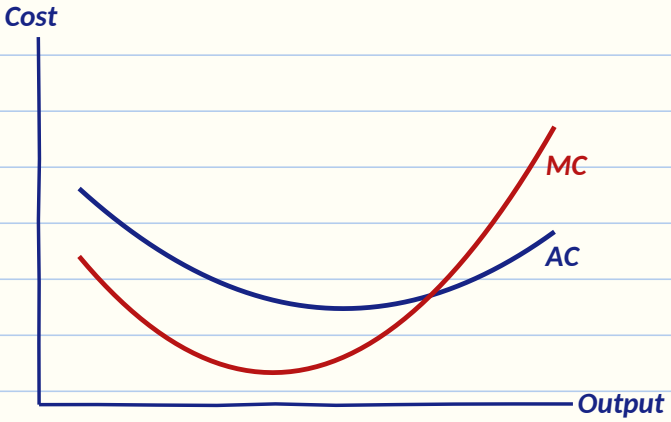
MC = change in TC / change in Q (= TCn - TCn-1)

- AC = AFC + AVC. AFC keeps falling as output rises (rectangular hyperbola).

4.4 Cost behaviour (short run)

- **TFC** is a horizontal line; **TVC** rises first slowly and then fast; **TC** starts from TFC level.

- **AC, AVC and MC are U-shaped** because of the law of variable proportions.
- **MC cuts AVC and AC at their minimum points.**



AC and MC are U-shaped.
MC cuts AC at its minimum point (law of variable proportions).

Cost schedule

Q	TFC	TVC	TC	AFC	AVC	AC	MC
0	100	0	100	-	-	-	-
1	100	60	160	100	60	160	60
2	100	100	200	50	50	100	40
3	100	150	250	33.3	50	83.3	50
4	100	240	340	25	60	85	90

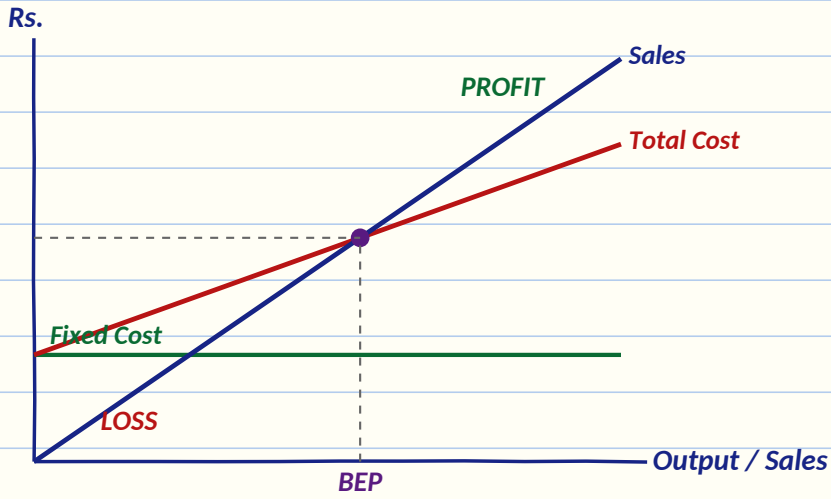
4.5 Long-run costs

- All costs are variable. The **LAC curve** is U-shaped (flatter) and is an **envelope curve** of short-run AC curves. **Economies of scale** first reduce LAC; **diseconomies** later raise it.
- Economies: internal (technical, managerial, marketing, financial) and external (localisation, information).

5. Break-Even Analysis (BEA)

5.1 Meaning

BEA (Cost-Volume-Profit analysis) studies the relation between **cost, volume and profit**. The **break-even point (BEP)** is the level of output / sales at which **total revenue = total cost** (no profit, no loss).



Left of BEP = loss

Right of BEP = profit

At BEP: Sales = Total Cost

(no profit, no loss)

5.2 Assumptions

- *Costs can be separated into fixed and variable; fixed cost is constant; variable cost per unit is constant.*
- *Selling price per unit is constant; production = sales; no change in technology or product mix.*

5.3 Key terms and formulas

Contribution per unit = Selling price (S) - Variable cost per unit (V)

BEP (units) = Fixed cost / (S - V)

BEP (Rs.) = Fixed cost / P/V ratio ; P/V ratio = Contribution / Sales

Margin of Safety = Actual sales - BEP sales ; MOS % = MOS / Actual sales

Sales for target profit = (Fixed cost + Desired profit) / Contribution per unit

5.4 Uses and advantages

- *Finds the **safe level** of output; helps in **price fixation**, product mix decision and cost control.*
- *Shows the effect of changes in cost and volume on **profit**; helps in make-or-buy and shutdown decisions.*

Limitations

- *Based on unrealistic assumptions (linear costs); useful only in the short run; ignores demand and competition; multi-product firms make it complex.*

6. Solved Problems on BEP

Problem 1

A firm has fixed cost Rs. 60,000; selling price Rs. 20 per unit; variable cost Rs. 12 per unit. Find (a) BEP in units, (b) BEP in Rs., (c) P/V ratio.

- Contribution per unit = 20 - 12 = **Rs. 8**.

(a) BEP units = 60,000 / 8 = 7,500 units

(b) BEP Rs. = 7,500 x 20 = Rs. 1,50,000

(c) P/V ratio = 8 / 20 = 40% (check: 60,000 / 0.40 = 1,50,000)

Problem 2 (profit and margin of safety)

Using the above data, actual sales are 10,000 units. Find profit and margin of safety.

- Profit = (Actual - BEP units) x contribution = (10,000 - 7,500) x 8 = **Rs. 20,000**.
- Check: Contribution = 10,000 x 8 = 80,000; minus fixed cost 60,000 = Rs. 20,000.
- MOS = 2,500 units (Rs. 50,000); MOS % = 2,500 / 10,000 = **25%**.

Problem 3 (target profit)

Fixed cost Rs. 1,00,000; S = Rs. 50; V = Rs. 30. How many units are needed to earn a profit of Rs. 40,000?

Units = (1,00,000 + 40,000) / (50 - 30) = 1,40,000 / 20 = 7,000 units

Problem 4 (change in price)

Fixed cost = Rs. 90,000, V = Rs. 15, S = Rs. 25. If the price falls to Rs. 22, find the new BEP.

- Old BEP = 90,000 / 10 = 9,000 units. New contribution = 22 - 15 = 7.

New BEP = 90,000 / 7 = 12,857 units (approximately)

- A price cut of Rs. 3 raises BEP by about 3,857 units.

Problem 5 (two products and profit)

Sales Rs. 5,00,000; variable cost Rs. 3,00,000; fixed cost Rs. 1,20,000. Find P/V ratio, BEP and profit.

- Contribution = 5,00,000 - 3,00,000 = 2,00,000; P/V = 2,00,000 / 5,00,000 = **40%**.

BEP (Rs.) = 1,20,000 / 0.40 = Rs. 3,00,000

Profit = Contribution - Fixed cost = 2,00,000 - 1,20,000 = Rs. 80,000

7. More Concepts for Exams

7.1 Economies and Diseconomies of Scale

Internal economies	Meaning
Technical	Large machines, specialisation, better technology, by-product use
Managerial	Specialist managers, division of work, decentralisation
Marketing	Bulk advertising, own sales network, brand image
Financial	Cheaper loans, better credit, easy share issue
Risk-bearing	Diversification of products and markets
Labour	Skilled workers, training, welfare facilities

- **External economies**: arise to all firms of an industry: localisation (cluster of suppliers), better transport, trained labour pool, information services, by-product industries.
- **Diseconomies**: too much size causes management inefficiency, labour disputes, shortage of raw materials, higher transport cost.

7.2 Ridge lines and Isoclines

- **Isocline** : line joining points of equal MRTS on different isoquants.
- **Ridge lines** : boundary lines of the economic region of production; outside them MP of a factor is negative.
- The firm produces in the **economic region** between ridge lines, where both inputs have positive marginal products.

7.3 Relationship between AC and MC

- When MC < AC, AC falls; when MC > AC, AC rises; when MC = AC, AC is at its minimum.
- Like a batting average: a new score below the average pulls it down; above the average pushes it up.

7.4 Classification of costs: Fixed vs Variable

Basis	Fixed cost	Variable cost
Behaviour	Constant at all levels of output	Changes with output
At zero output	Still exists	Zero
Per unit	Falls as output rises	Constant per unit
Examples	Rent, insurance, salaries, depreciation	Raw materials, fuel, piece-rate wages

7.5 Short-run vs Long-run cost curves

- SAC curves are U-shaped because of **law of variable proportions**.

- LAC is U-shaped because of **returns to scale** (economies then diseconomies).
- LAC is the **envelope** of the SAC curves: it touches each SAC curve at one point only.

7.6 Cost-Volume-Profit extras

- **Angle of incidence**: the angle at which the sales line cuts the total cost line at BEP. A large angle means a high profit rate beyond BEP.
- **Contribution** = Sales - Variable cost = Fixed cost + Profit.
- Reducing BEP: reduce fixed cost, reduce variable cost, raise price, change product mix.
- **Shutdown point**: firm should continue in the short run as long as price covers AVC (contribution is positive).

7.7 Differences quick table

Isoquant	Indifference curve
Shows equal output (cardinal, measurable)	Shows equal satisfaction (ordinal)
Used in production theory	Used in consumer theory
Inputs: labour and capital	Goods: X and Y
Slope = MRTS	Slope = MRS

8. More Solved Problems

Problem 6: Cost schedule

TFC = Rs. 200. TVC for 1, 2, 3 and 4 units is Rs. 100, 180, 240 and 340. Find TC, AC and MC for each.

Q	TC	AC	MC
1	300	300	100
2	380	190	80
3	440	146.7	60
4	540	135	100

- MC for the 1st unit = 300 - 200 = 100 (TC at zero output is TFC = 200). MC rises again after the 3rd unit, so diminishing returns set in.

Problem 7: Profit at a given price

TFC = Rs. 50,000; AVC = Rs. 40; price = Rs. 65; output 4,000 units. Find BEP and profit.

- Contribution per unit = 65 - 40 = 25.

BEP = 50,000 / 25 = 2,000 units

Profit = (4,000 - 2,000) x 25 = Rs. 50,000

Problem 8: Make a decision between two plans

Plan A: fixed cost Rs. 40,000, V = Rs. 8. Plan B: fixed cost Rs. 70,000, V = Rs. 5. Find the output where both plans cost the same.

- Set TC equal: 40,000 + 8Q = 70,000 + 5Q.

3Q = 30,000, so Q = 10,000 units

- Below 10,000 units choose Plan A (lower fixed cost); above 10,000 units choose Plan B (lower variable cost).

Problem 9: Required price

A firm sells 5,000 units. Fixed cost Rs. 80,000, variable cost Rs. 24 per unit. Find the selling price to break even.

At BEP: S x 5,000 = 80,000 + 24 x 5,000 = 2,00,000, so S = Rs. 40 per unit

Problem 10: Production function

A firm uses Q = 10L - L squared (capital fixed). Find the L that maximises output.

- MP = dQ/dL = 10 - 2L. Set MP = 0, so L = 5; maximum Q = 50 - 25 = 25.
- Beyond L = 5, MP is negative (Stage III), so a rational firm stops at or before 5 workers.

Problem 11: Least-cost check

MPL = 40, MPK = 100, w = Rs. 10, r = Rs. 20. Is the firm at the least-cost combination?

- MPL/w = 40/10 = 4; MPK/r = 100/20 = 5. They are unequal (4 < 5).
- Capital gives more output per rupee, so the firm should **use more capital and less labour** until MPL/w = MPK/r.

9. Format of long answers

Tip: In a 10-mark answer on any topic write: Definition, Features, Diagram, Explanation / Types, Merits and Limitations, and a short conclusion.

- For **law of variable proportions** : statement, assumptions, schedule, 3 stages with

diagram, causes, limitations.

- For **BEP** : meaning, assumptions, formulas, chart, uses, limitations, one numerical.

10. Quick Revision Sheet

Concept	Key point
Production function	Q = f(L, K); technical relation inputs and output
Short run law	Variable proportions: 3 stages; operate in Stage II
Long run	Returns to scale: increasing, constant, decreasing
Isoquant	Same output curve; convex, downward, non-intersecting
Isocost	Same cost line; slope = w / r
Least-cost point	MRTS = w / r (tangency)
Cost curves	AC, AVC, MC U-shaped; MC cuts AC at minimum
BEP	Fixed cost / (S - V)

Remember: AFC falls continuously; MC = change in TC; MC = AC at minimum AC; at BEP profit = 0.

Important Exam Questions

- Explain the production function. Distinguish between short run and long run. (10 M)
- Explain the law of variable proportions with a diagram. (10 M)
- What are isoquants and isocost lines? Explain the least-cost combination of inputs. (10 M)
- Explain the various cost concepts. How do costs behave with output? (10 M)
- What is break-even analysis? Explain assumptions, uses and limitations. (10 M)
- Problems on BEP, P/V ratio, margin of safety and target profit.